

[Niacin: Fact Sheet for Consumers \(original version\)](#)

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What is niacin and what does it do?

Niacin (also called vitamin B3) helps turn the food you eat into the energy you need. Niacin is important for the development and function of the cells in your body.

How much niacin do I need?

The amount of niacin you need depends on your age and sex. Average daily recommended amounts are listed below in milligrams (mg) of niacin equivalents (NE) (except for infants in their first 6 months).

The mg NE measure is used because your body can also make niacin from tryptophan, an amino acid in proteins. For example, when you eat turkey, which is high in tryptophan, some of this amino acid is converted to niacin in your liver. Using mg NE accounts for both the niacin you consume and the niacin your body makes from tryptophan. Infants in their first 6 months do not make much niacin from tryptophan.

Life Stage	Recommended Amount
Birth to 6 months	2 mg
Infants 7–12 months	4 mg NE
Children 1–3 years	6 mg NE
Children 4–8 years	8 mg NE
Children 9–13 years	12 mg NE
Teen boys 14–18 years	16 mg NE
Teen girls 14–18 years	14 mg NE
Adult men 19+ years	16 mg NE
Adult women 19+ years	14 mg NE
Pregnant teens and women	18 mg NE
Breastfeeding teens and women	17 mg NE

What foods provide niacin?

Niacin is found naturally in many foods and is added to some foods. You can get recommended amounts of niacin by eating a variety of foods, including the following:

- Animal foods, such as poultry, beef, pork, and fish
- Some types of nuts, legumes, and grains
- Enriched and fortified foods, such as many breads and cereals

What kinds of niacin dietary supplements are available?

Niacin is found in multivitamin/mineral supplements. It is also available in B-complex dietary supplements and supplements containing only niacin. The two main forms of niacin in dietary supplements are nicotinic acid and nicotinamide.

Niacin (in the form of nicotinic acid) is also available as a prescription medicine used to treat high blood cholesterol levels.

Am I getting enough niacin?

Most people in the United States get enough niacin from the foods they eat. Niacin deficiency is very rare in the United States. However, some people are more likely than others to have trouble getting enough niacin:

- Undernourished people with AIDS, alcohol use disorder, anorexia, inflammatory bowel disease, or liver cirrhosis
- People whose diet has too little iron, riboflavin, or vitamin B6; these nutrients are needed to convert tryptophan to niacin
- People with Hartnup disease, a rare genetic disorder
- People with carcinoid syndrome, a condition in which slow-growing tumors develop in the gastrointestinal tract

What happens if I don't get enough niacin?

You can develop niacin deficiency if you don't get enough niacin or tryptophan from the foods you eat. Severe niacin deficiency leads to a disease called pellagra. Pellagra, which is uncommon in developed countries, can have the following effects:

- Rough skin that turns red or brown in the sun
- A bright red tongue
- Vomiting, constipation, or diarrhea
- Depression
- Headaches
- Extreme tiredness
- Aggressive, paranoid, or suicidal behavior
- Hallucinations, apathy, loss of memory

In its final stages, pellagra leads to loss of appetite followed by death.

What are some effects of niacin on health?

Scientists are studying niacin to better understand how it affects health. Here is an example of what this research has shown.

Cardiovascular disease

Scientists have studied the use of large doses of niacin in the form of nicotinic acid to help reduce the risk of heart attack and stroke in people with atherosclerosis. They found that prescription-strength nicotinic acid

(more than 100 times the recommended dietary allowance) can lower blood levels of LDL (bad) cholesterol, raise levels of HDL (good) cholesterol, and lower levels of triglycerides. However, these favorable effects on blood lipids (fats) don't affect the risk of having a cardiovascular event, such as heart attack, sudden cardiac death, or stroke. In addition, experts do not recommend high doses of nicotinic acid for people taking a statin medication.

Your health care provider should approve and supervise any use of very high doses of nicotinic acid (in the thousands of milligrams) to treat atherosclerosis.

Can niacin be harmful?

The niacin that food and beverages naturally contain is safe. However, dietary supplements with 30 mg or more of nicotinic acid can make the skin on your face, arms, and chest turn red and burn, tingle, and itch. This can also lead to headaches, rashes, and dizziness.

If you take nicotinic acid as a medication in doses of 1,000 or more mg/day, it can cause more severe side effects. These include:

- Low blood pressure (which can increase the risk of falls)
- Extreme tiredness
- High blood sugar levels
- Nausea, heartburn, and abdominal pain
- Blurred or impaired vision and fluid buildup in the eyes

Long-term treatment, especially with extended-release forms of nicotinic acid, can cause liver problems, including hepatitis and liver failure.

Niacin in the form of nicotinamide has fewer side effects than nicotinic acid. However, at high doses of 500 mg/day or more, nicotinamide can cause diarrhea and easy bruising and can increase bleeding from wounds. Even higher doses of 3,000 mg/day or more can cause nausea, vomiting, and liver damage.

The daily upper limits for niacin from dietary supplements are listed below.

Ages	Upper Limit
Birth to 12 months	Not established
Children 1–3 years	10 mg
Children 4–8 years	15 mg
Children 9–13 years	20 mg
Teens 14–18 years	30 mg
Adults 19+ years	35 mg

Does niacin interact with medications or other dietary supplements?

Niacin dietary supplements can interact or interfere with certain medicines that you take, and some medicines can lower niacin levels in your body. Here are some examples:

- Tuberculosis drugs (such as isoniazid and pyrazinamide) interfere with the body's ability to convert tryptophan to niacin. This interference can lower niacin levels in your body.
- High doses of nicotinic acid (1,500 mg/day or more) can raise blood sugar levels and interfere with the effectiveness of diabetes medications. These doses can even raise blood sugar levels in people who don't have diabetes.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your

medicines. They can also tell you if the medicines might interfere with how your body absorbs, uses, or breaks down niacin and other nutrients.

Niacin and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about niacin?

- For general information on niacin
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Niacin](#)
- For more information on food sources of niacin
 - ODS Health Professional Fact Sheet on [Niacin](#)
 - USDA's [FoodData Central](#)
 - Nutrient List for niacin (listed by [food](#) or by [nutrient content](#)), USDA
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#).

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

adverse event

An unwanted medical problem that occurs during treatment. Adverse events may be unrelated to the treatment or they may be caused by the therapy or procedure. For example, an adverse event may be caused by the toxic effects of a particular drug or dietary supplement or by an interaction with another therapy. Also called adverse effect and side effect.

amino acid

A chemical building block of protein.

anorexia

Not feeling hungry or wanting to eat food to an unusual extent.

apathy

Having no emotion, interests, or concerns.

atherosclerosis

A condition in which certain blood vessels (arteries) are clogged and have hardened. Atherosclerosis is caused by fat and cholesterol deposits (plaque) that block blood flow to certain parts of the body. It increases a person's chance (risk) of having a heart attack, stroke, kidney damage, or other diseases involving the arteries.

blood sugar

The main source of energy used by the body's cells. Blood sugar comes from food and is made by the liver, and is carried to the cells through the bloodstream. Also called blood glucose.

cardiovascular event

A heart attack, stroke, or other occurrence that damages the heart or blood vessels.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

cholesterol

A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

cirrhosis

A condition in which damaged liver cells are replaced with scar tissue, making it progressively more difficult for the liver to function properly.

constipation

A condition in which stool becomes hard, dry, and difficult to pass and bowel movements happen infrequently. Other symptoms may include painful bowel movements and feeling bloated, uncomfortable, and sluggish.

consume

To eat or drink.

deficiency

An amount that is not enough; a shortage.

depression

A serious medical condition that can interfere with an individual's ability to work, study, sleep, and eat. Symptoms include ongoing feelings of sadness and despair, loss of energy, and loss of interest or pleasure in activities that were once enjoyed. A person who has depression may experience changes in eating or sleeping habits, and have thoughts of death or suicide.

diabetes

A disease in which blood sugar (glucose) levels are high because the body is unable to use glucose properly. Diabetes occurs when the body does not make enough insulin, which helps the cells use glucose, or when the body no longer responds to insulin.

diarrhea

Loose, watery stools.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

effectiveness

In medicine, the ability of an intervention (for example, a drug, surgery, or a dietary supplement) to produce the desired beneficial effect under the usual conditions of care by a health care provider.

enriched

When certain nutrients (such as vitamins and minerals) have been added to a food product to replace nutrients that may be lost during processing or storage. For example, white flour is enriched with thiamin, riboflavin, niacin, and iron because those nutrients are lost when grain is made into flour. Folic acid is also added to enriched flour.

extended-release

In medicine, pills and capsules that release a drug in the body slowly over time.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gastrointestinal tract

The large, muscular tube that extends from the mouth to the anus, where the movement of muscles and release of hormones and enzymes digest food.

genetic disorder

A disease or disorder caused by an alteration or variation (mutation) in a gene or group of genes in the cells of an individual. Examples of genetic disorders include breast cancer, cystic fibrosis, Parkinson's disease, and celiac disease. They can be inherited or can occur without a known cause.

heart attack

The blockage of an artery supplying blood and oxygen to the heart, resulting in the damage or death of a section of heart muscle.

hepatitis

A group of diseases in which the liver becomes enlarged and inflamed, causing fever, nausea, vomiting, abdominal pain, and dark urine.

infant

A child younger than 12 months old.

inflammatory bowel disease

IBD. Long-lasting (chronic) problems that cause irritation and ulcers in the digestive tract. The most common disorders are ulcerative colitis and Crohn's disease.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

iron

In nutrition, a mineral the body needs to make red blood cells, proteins, and enzymes; and for the control of cell growth and cell specialization. Iron is found in some foods, including red meats, fish, poultry, lentils, and beans.

legume

Dried beans and peas, including kidney beans, pinto beans, black beans, navy beans, lima beans, black-eyed peas, garbanzo beans (chickpeas), split peas, and lentils. Legumes are good sources of protein, iron, zinc, dietary fiber, folate, and potassium.

liver

A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

milligram

mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nausea

The uneasy feeling of having an urge to throw up (vomit).

niacin

A nutrient that is needed by the body to make energy from food, and for important cellular functions. Niacin is found in some foods, including poultry, beef, pork, fish, nuts, legumes, whole grains, and enriched or fortified breads and cereals. In medicine it is used as a drug to lower high blood cholesterol levels. It is one of the B vitamins. Also called nicotinamide and nicotinic acid.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by

evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

poultry

Birds that are raised for eggs or meat, including chickens, turkeys, ducks, and geese.

prescription

A written order from a health care provider for medicine, therapy, or tests.

protein

A molecule made up of amino acids that the body needs for good health. Proteins are the basis of body structures such as skin and muscle, and substances such as enzymes and antibodies.

Recommended Dietary Allowance

RDA. The average amount of a nutrient a healthy person should get each day. RDAs vary by age, gender and whether a woman is pregnant or breastfeeding. For example, the RDA for vitamin C is 80 mg a day for a pregnant teenager and 90 mg a day for men. RDAs are developed by the Food and Nutrition Board at the Institute of Medicine of the National Academies.

riboflavin

A nutrient that is needed by the body for growth, cell function, and to make energy from food. It works together with other B vitamins, and acts as an antioxidant to protect cells from free radical damage. Riboflavin is found in some foods, including liver, mushrooms, spinach and leafy green vegetables, milk and dairy products, eggs, whole grains, enriched breads and cereals, and lean meats. Also called vitamin B₂.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

statin

A drug used to treat high cholesterol. Statins lower the amount of cholesterol and certain fats in the blood.

stroke

A loss of blood flow to part of the brain. Strokes are caused by blood clots or broken blood vessels in the brain, and result in damage to a section of brain tissue. Symptoms include dizziness, numbness, weakness on one side of the body, and problems with talking or understanding language. The chance (risk) of stroke is increased by high blood pressure, older age, smoking, diabetes, high cholesterol, heart disease, a family history of stroke, and a build-up of fatty material inside the coronary arteries (atherosclerosis). See also NIH publication: Know Stroke. Know the Signs. Act in Time.

<http://www.ninds.nih.gov/disorders/stroke/knowstroke.htm>

sudden cardiac death

A condition in which the heart suddenly stops beating. This usually causes death if not treated within minutes.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

triglyceride

A type of fat found in your blood. When you eat more than you need, your body turns the excess calories into triglycerides. High blood levels of triglycerides can increase your risk of heart disease, heart attack, and stroke.

tuberculosis

A disease caused by a specific type of bacteria that spreads from one person to another through the air. Tuberculosis can affect many parts of the body, but most often affects the lungs. A person may not have symptoms of tuberculosis for years, but they may appear when the patient becomes ill with a serious condition like diabetes, AIDS, or cancer. Tuberculosis can usually be treated and cured with antibiotics. Also called TB.

tumor

An abnormal growth of tissue that forms when cells divide more often than they should. Tumors may be benign (not cancer) or malignant (cancer).

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin B₆

A group of water-soluble chemical compounds, including pyridoxine, pyridoxal, and pyridoxamine. It is involved in protein metabolism, is needed for the nervous system and immune system to work efficiently, and is needed to make hemoglobin (a molecule within red blood cells that carries oxygen to tissues). It also helps maintain blood glucose (sugar) within a normal range. For more information see the NIH Office of Dietary Supplements Vitamin B₆ fact sheet.

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